

Assignment # 02

Signals and Systems

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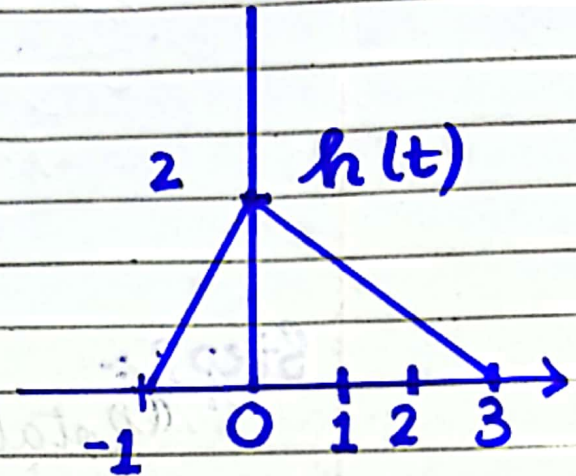
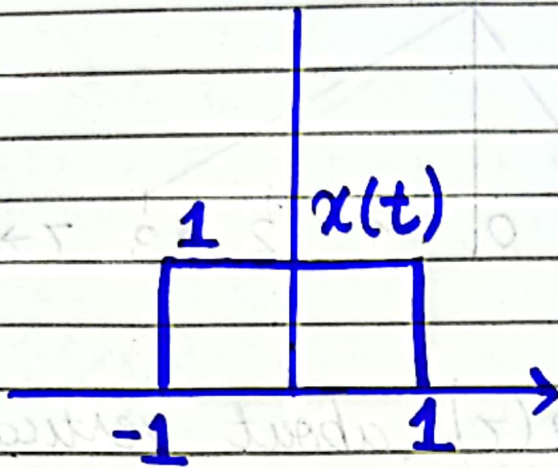
Section:

BEE-4A

Submitted To:

Sir Shaikh Muhammad Sajid

Questions # 01

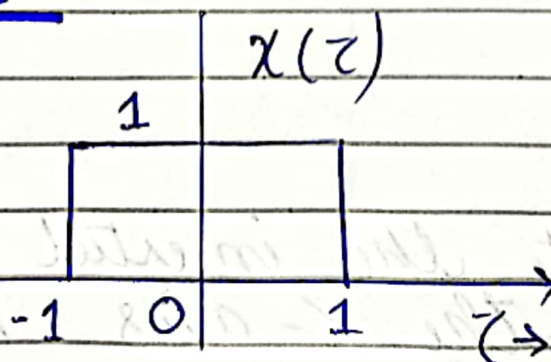


$$y(t) = x(t) * h(t)$$

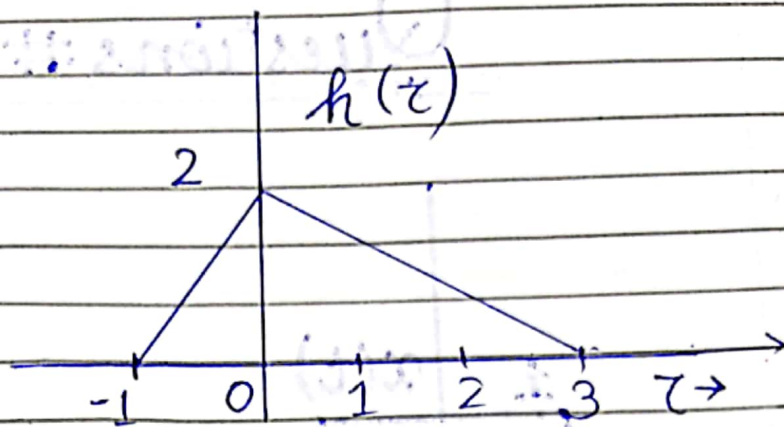
Use graphical convolution techniques to sketch $y(t)$

Solution:-

Step 1:-

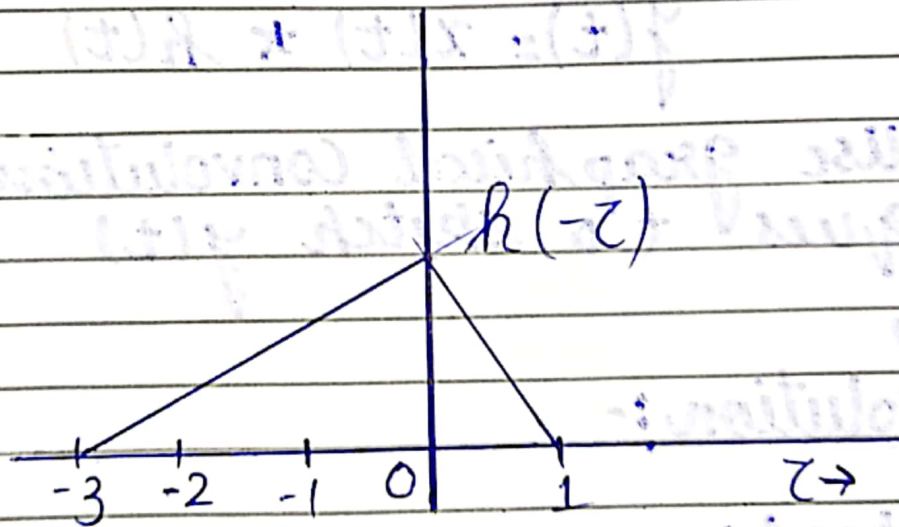


" $x(z)$ will remain fixed"



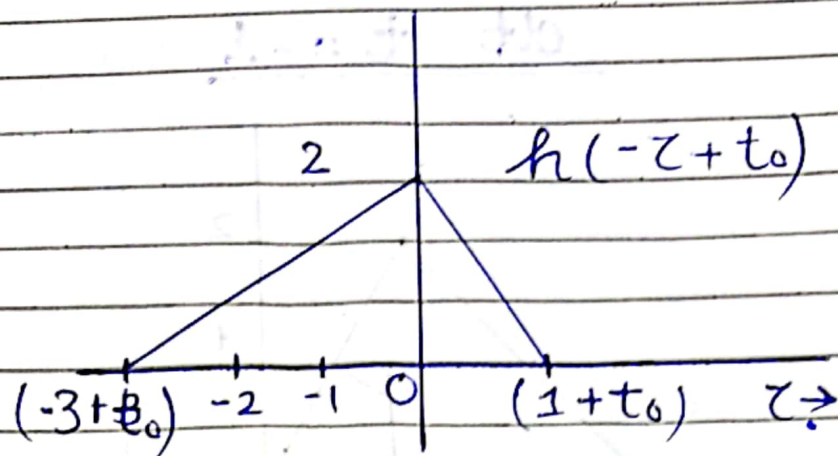
Step 2:-

"Rotate $h(z)$ about vertical axis"



Step 3:-

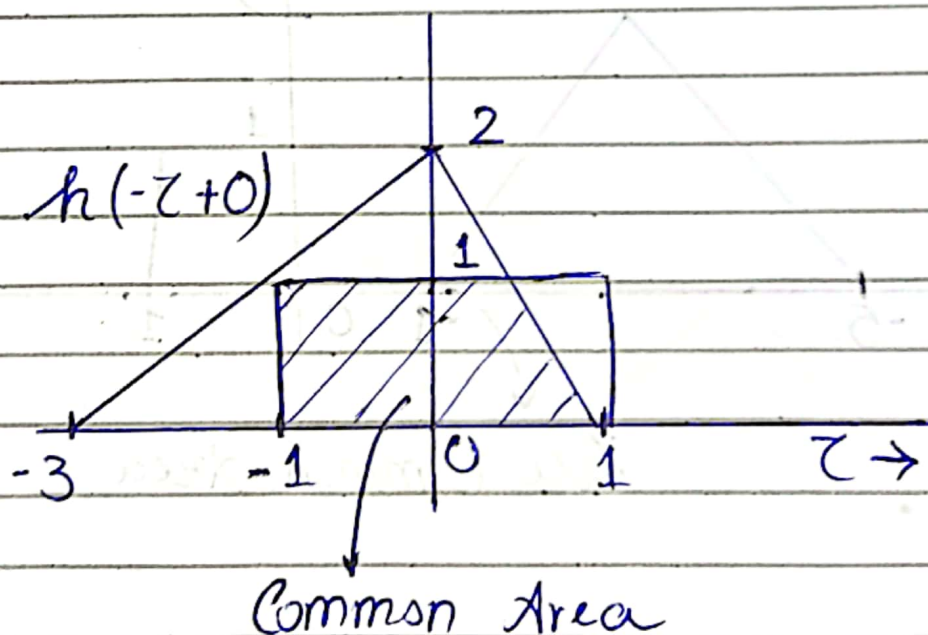
Shift the inverted frame along the z -axis by " t_0 "



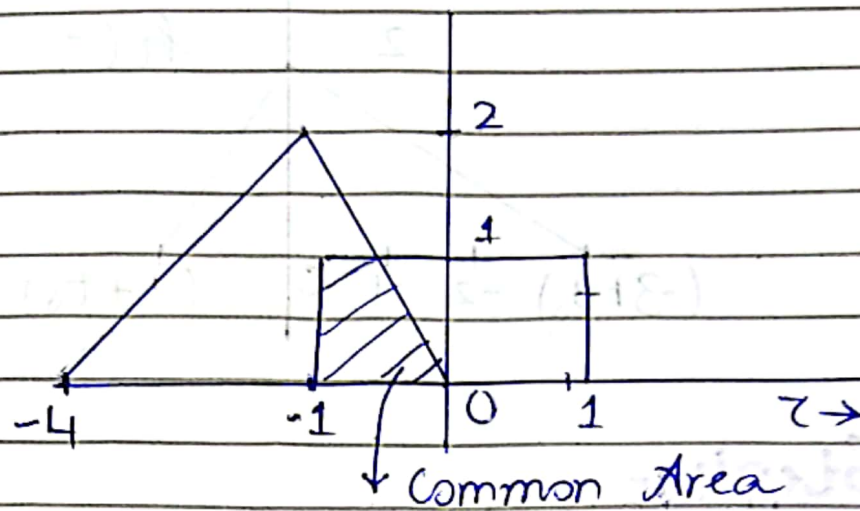
Step 4:-

→ Slide the $h(-z + t_0)$ near $x(z)$ at different " t_0 's values"

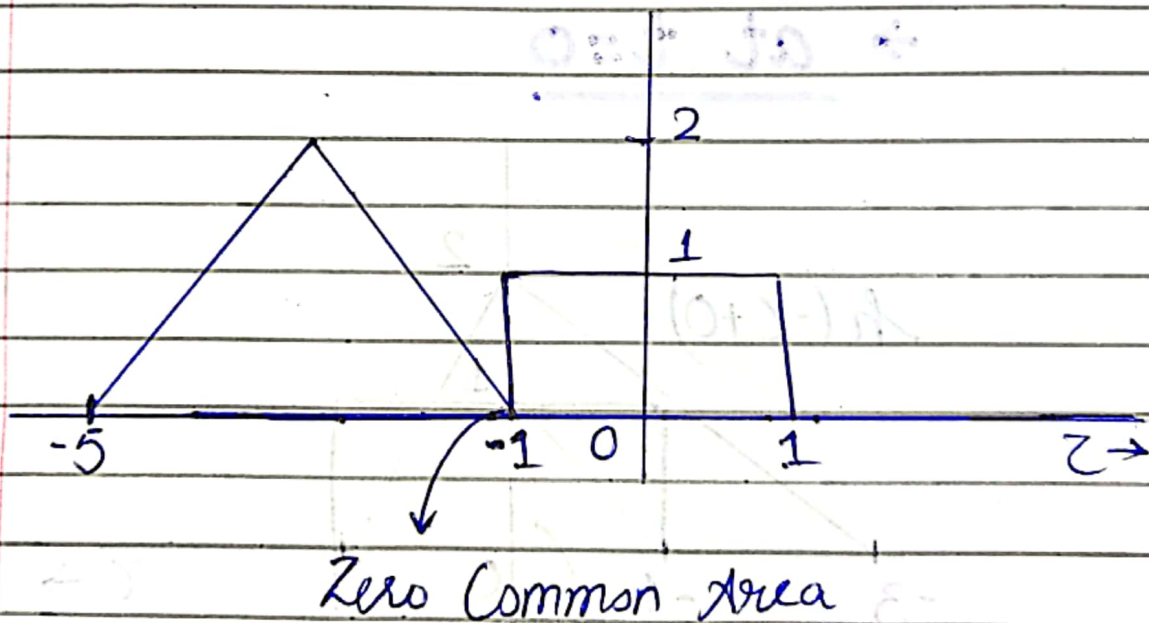
→ at $t = 0$



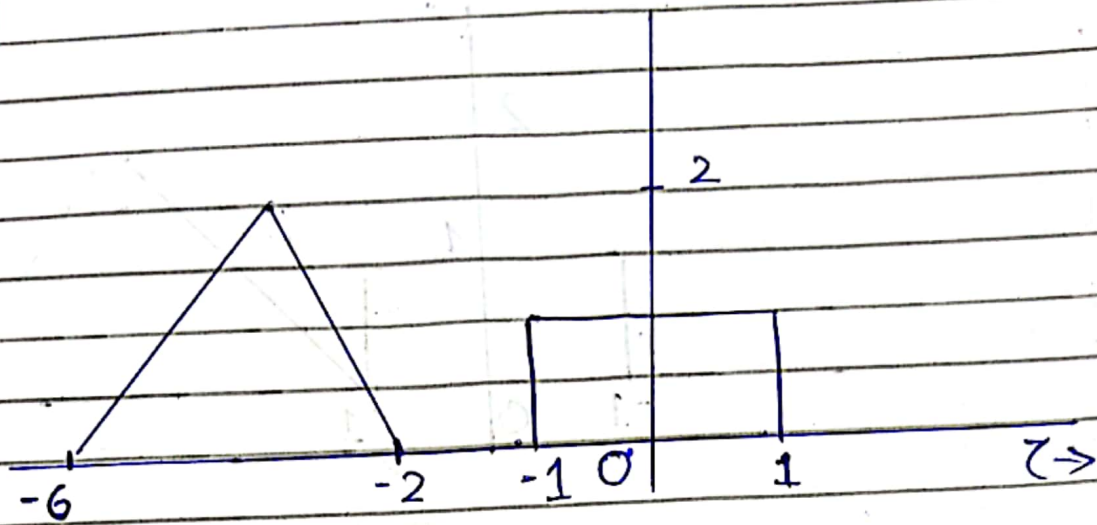
at $t = -1$



at $t = -2$

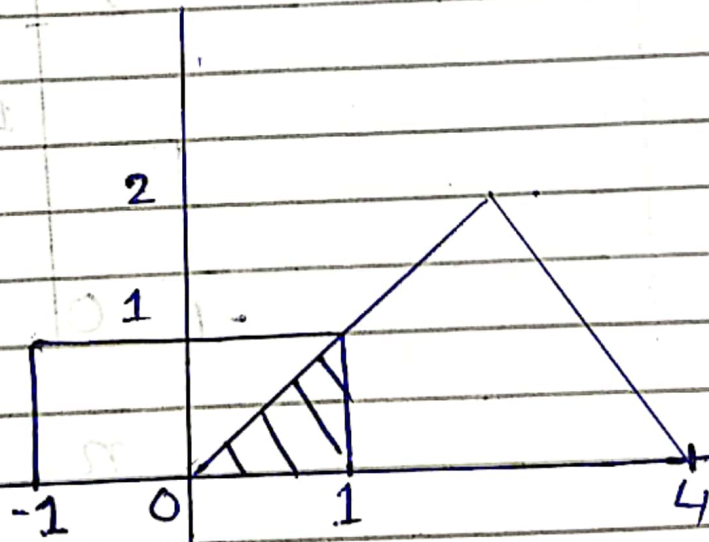


at $t = -3$

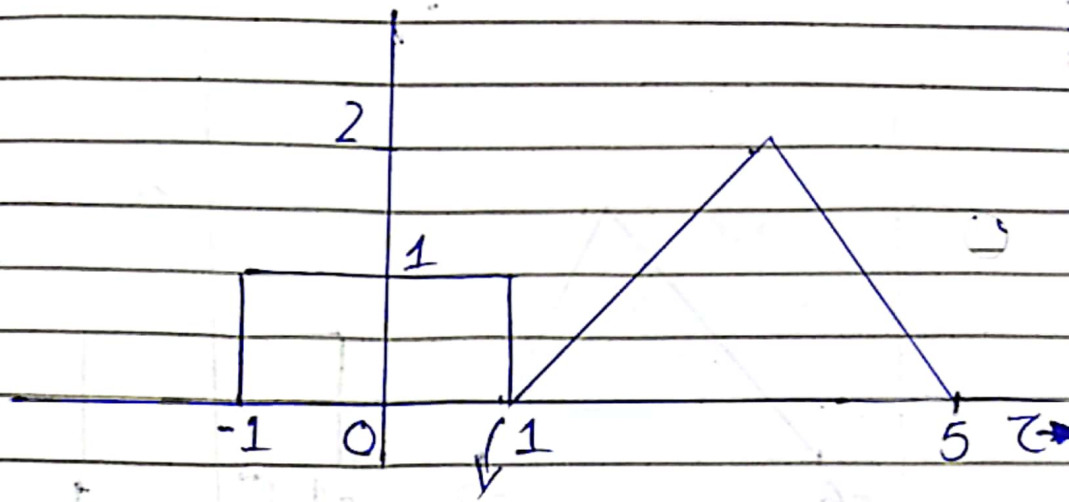


No Common Area

at $t = 3$

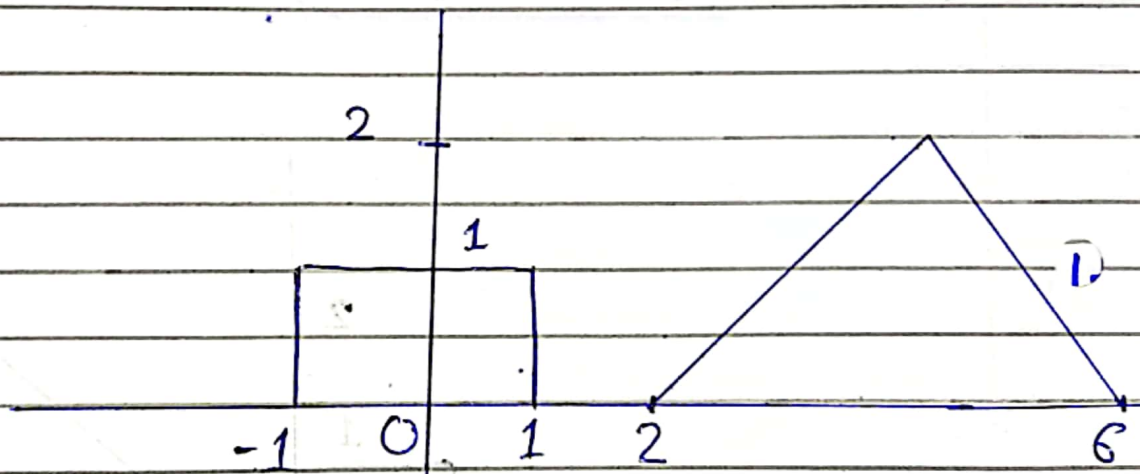


at $t=4$



0 common area

at $t=5$



no common area

Therefore;

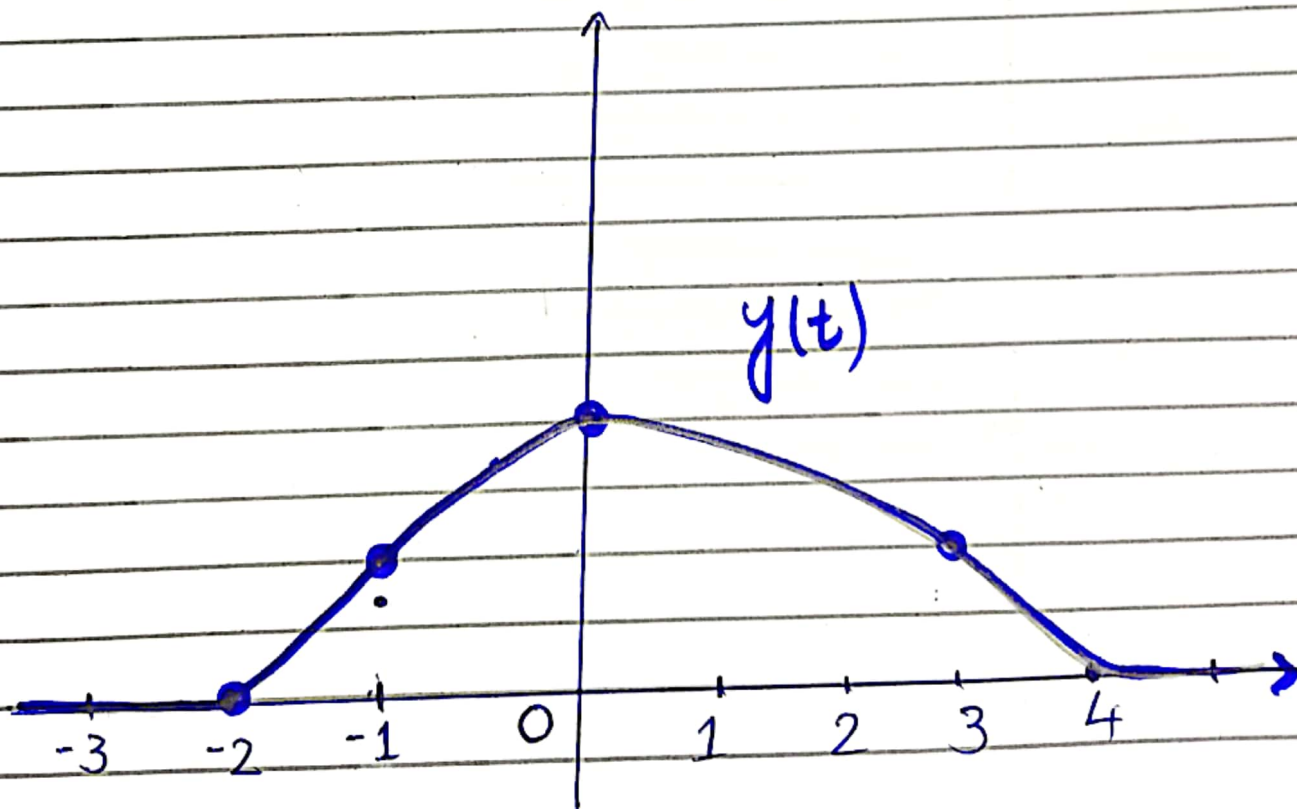
→ for $t_0 \leq -2$ i.e. -3, -4

there will be no common area

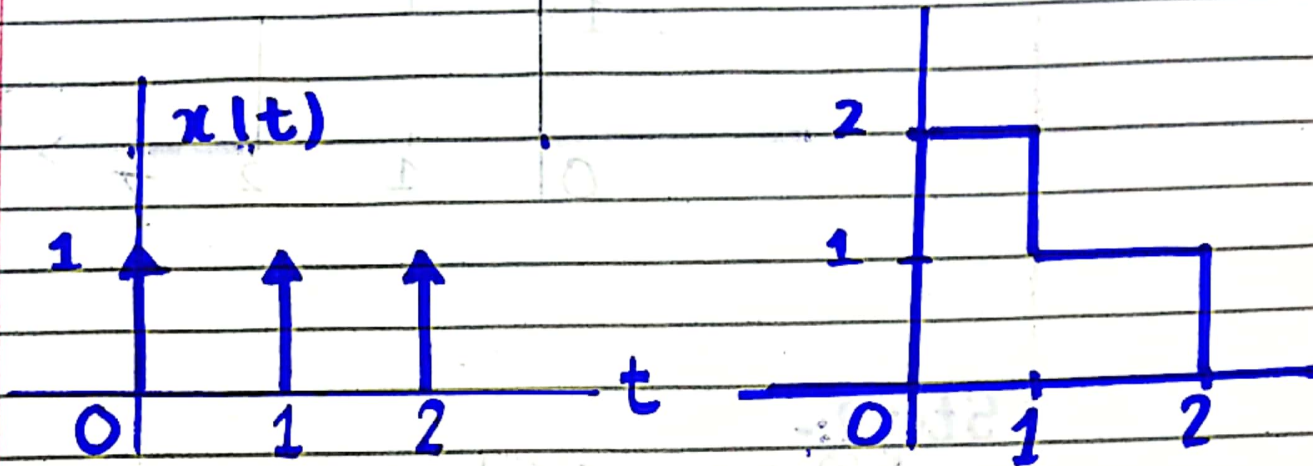
→ for $t_0 \geq 4$ i.e. 5, 6 also
there will be no common
area

Common Area Range:-

$$-1 \leq t_0 < 4$$



Question #02

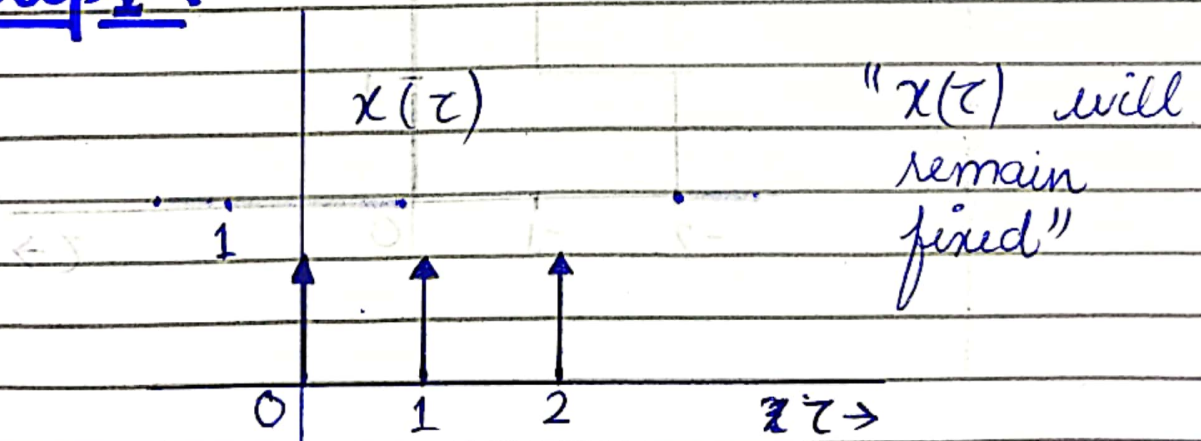


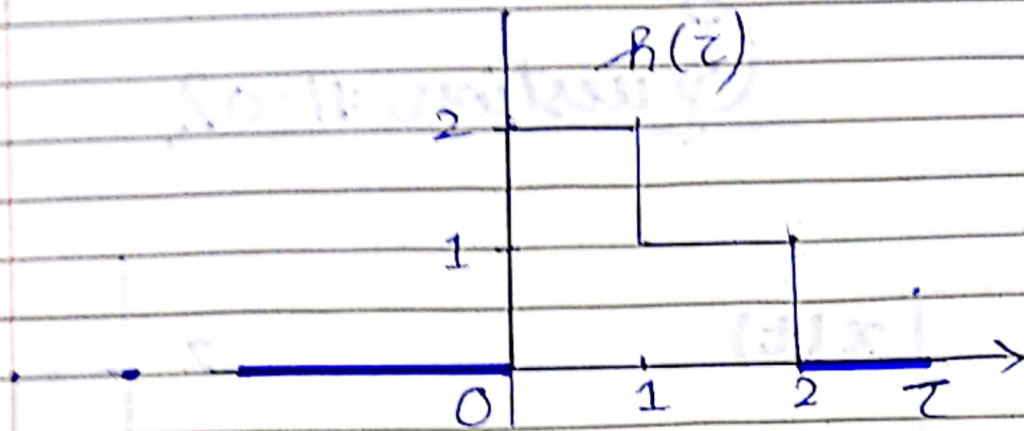
$$y(t) = x(t) * h(t)$$

Graphical convolution :-

Solution :-

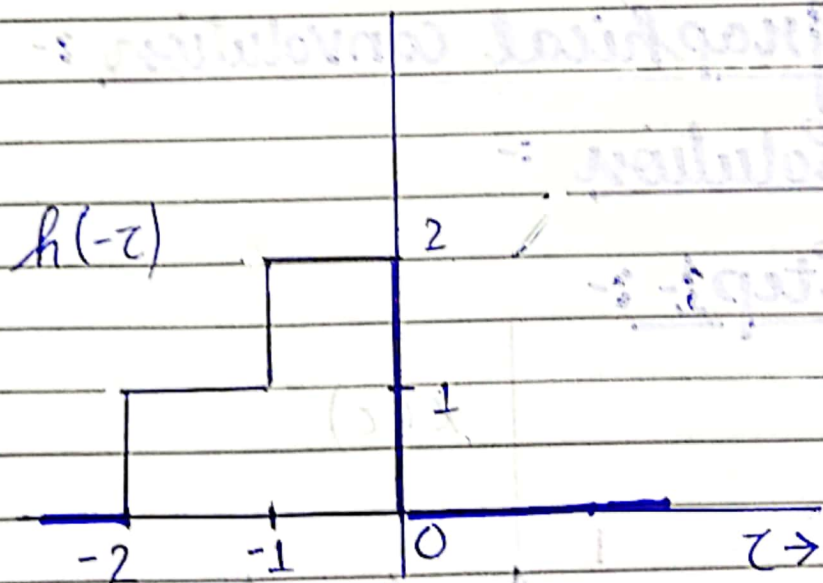
Step 1 :-





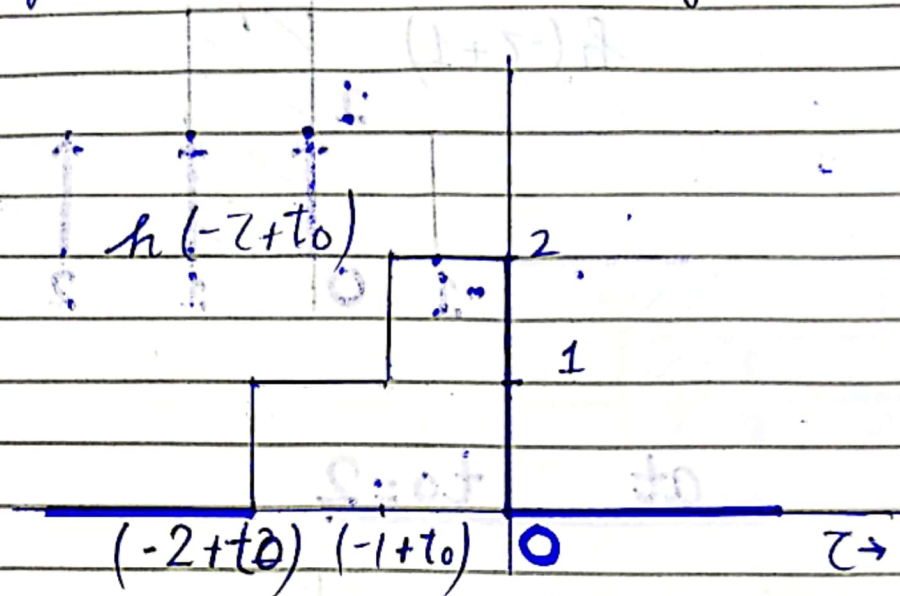
Step 2:-

"Rotate $h(\tau)$ about vertical axis"



Step 3:-

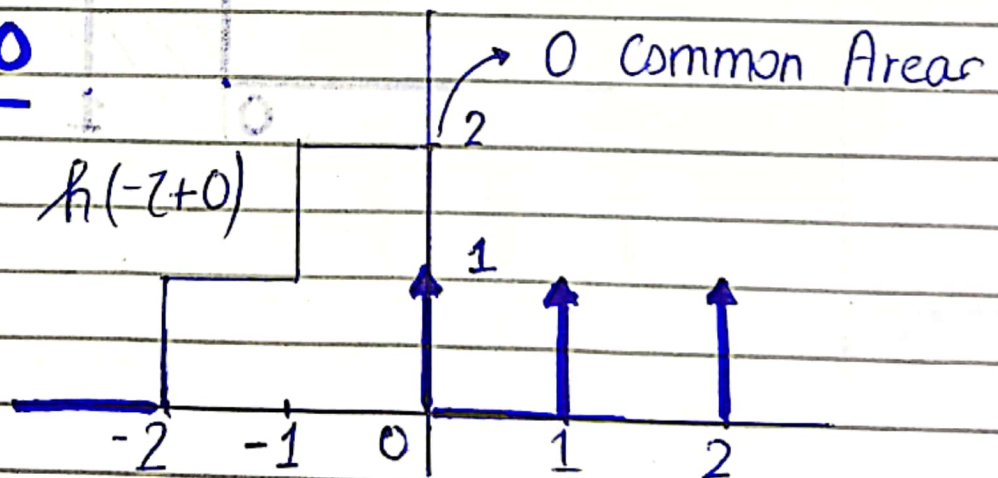
Shift the inverted frame $h(-z)$ along the z -axis by " t_0 "



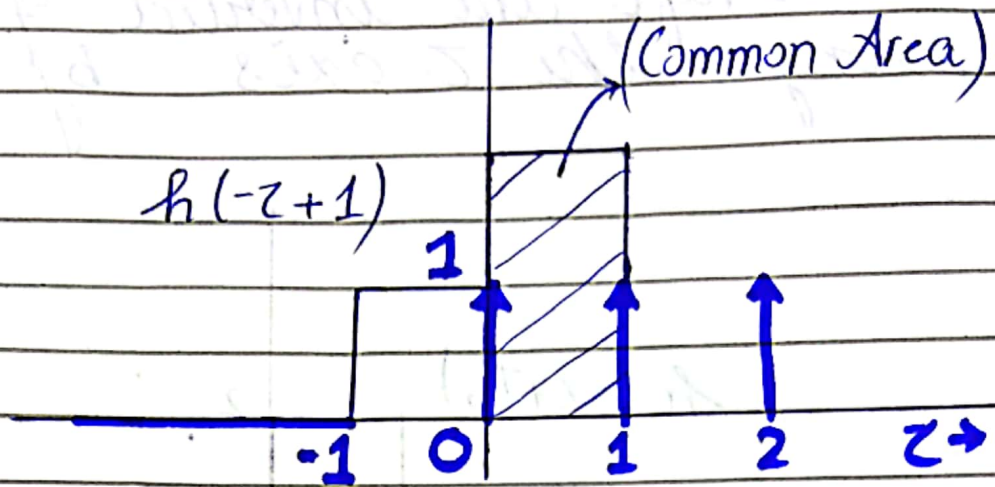
Step 4:-

Slide the $h(-z+t_0)$ near $x(z)$ at different " t_0 's" values.

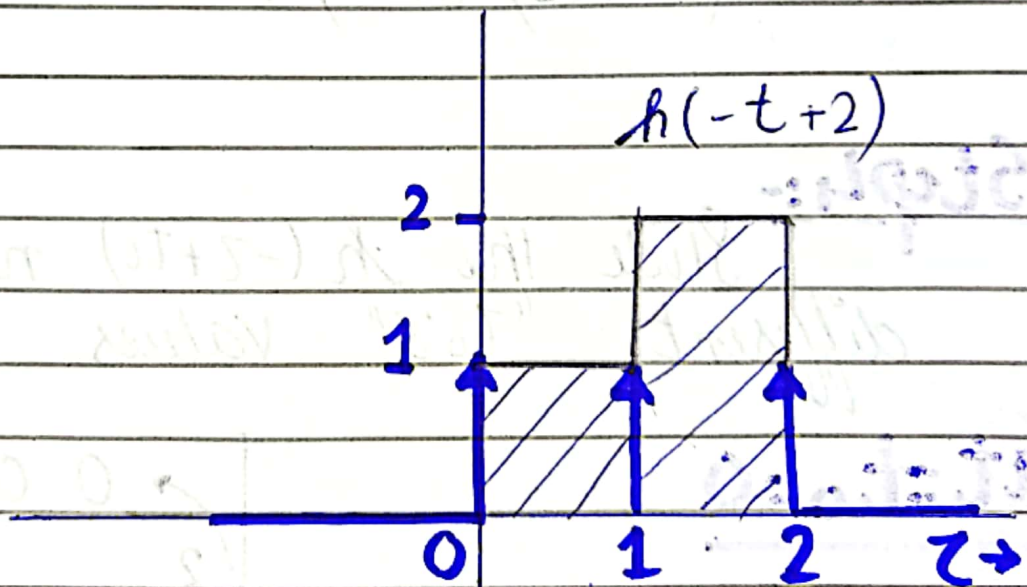
at $t_0=0$



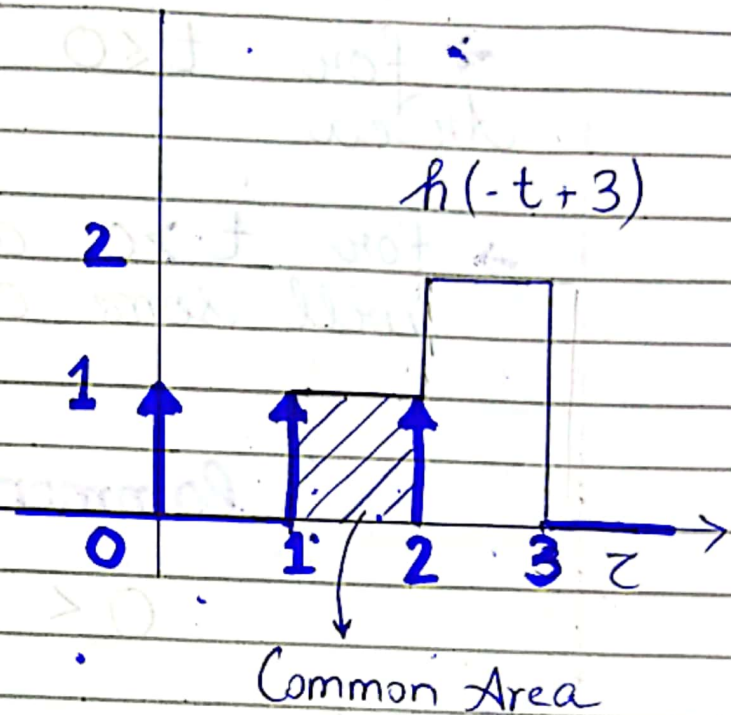
at $t_0 = 1$



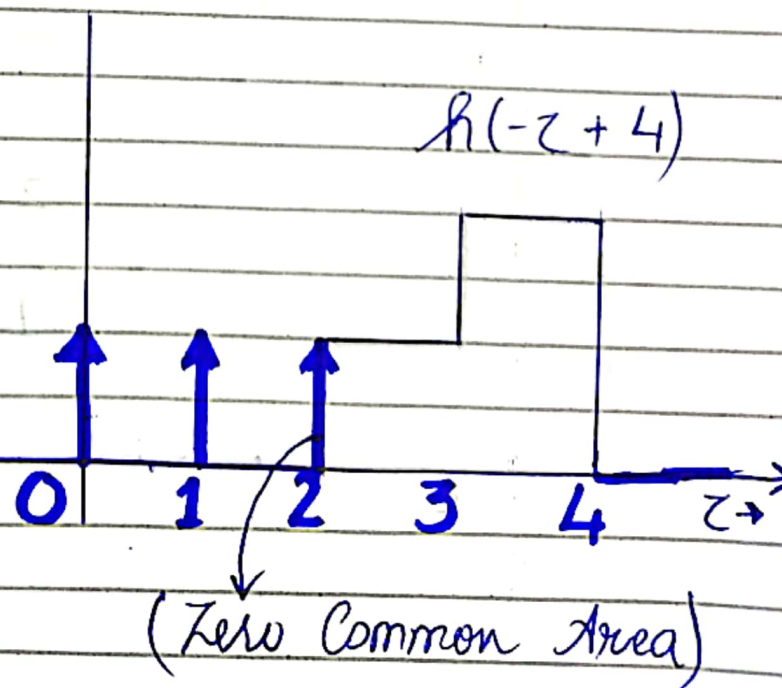
at $t_0 = 2$



at $t_0 = 3$



at $t_0 = 4$



Conclusion :-

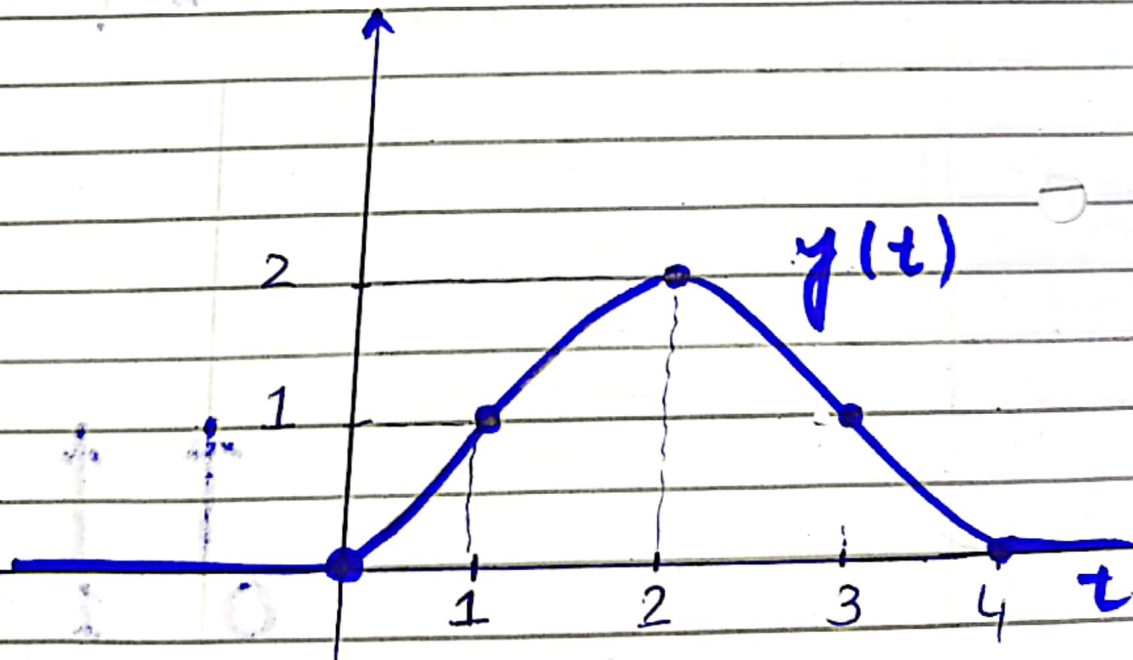
→ for $t \leq 0$ there is no common Area

→ for $t > 0$ and $t \leq 3$ there will some common Area

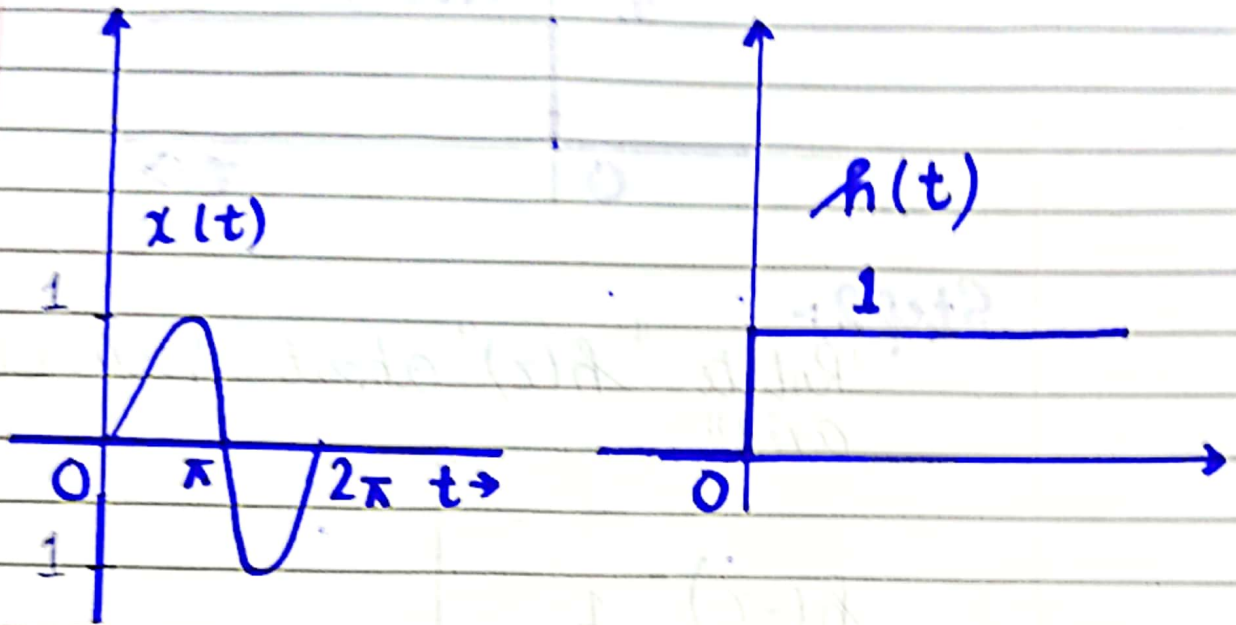
Common Area Range:-

$$0 < t \leq 3$$

Therefore



Question # 03

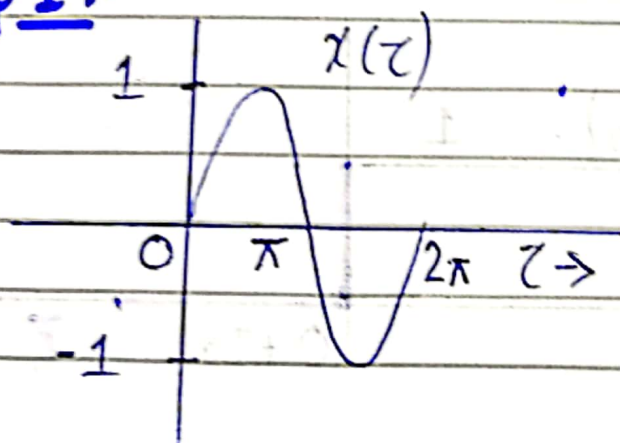


$$y(t) = x(t) * h(t)$$

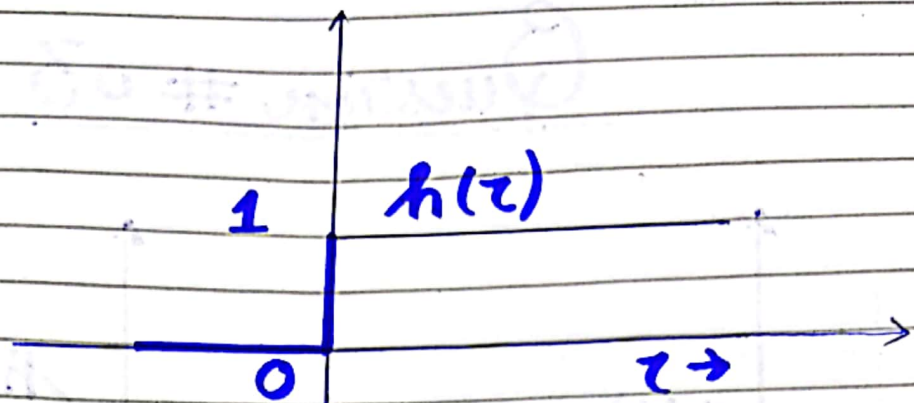
graphical convolution

Solution:-

Step 1:-

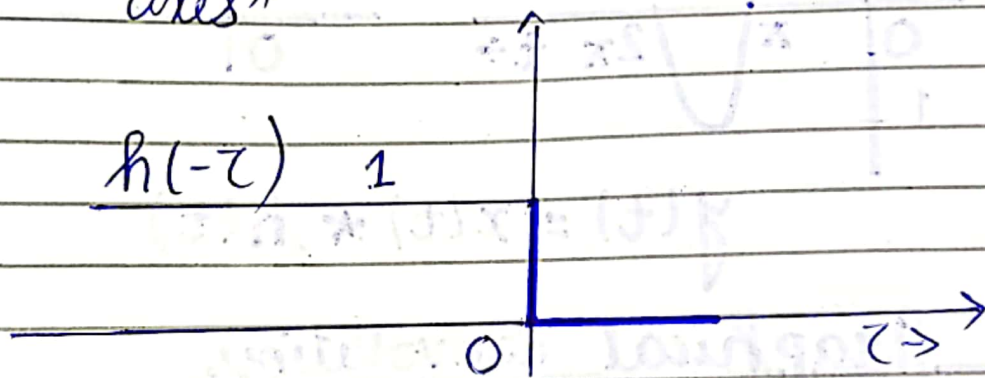


" $x(\tau)$ will remain fixed"



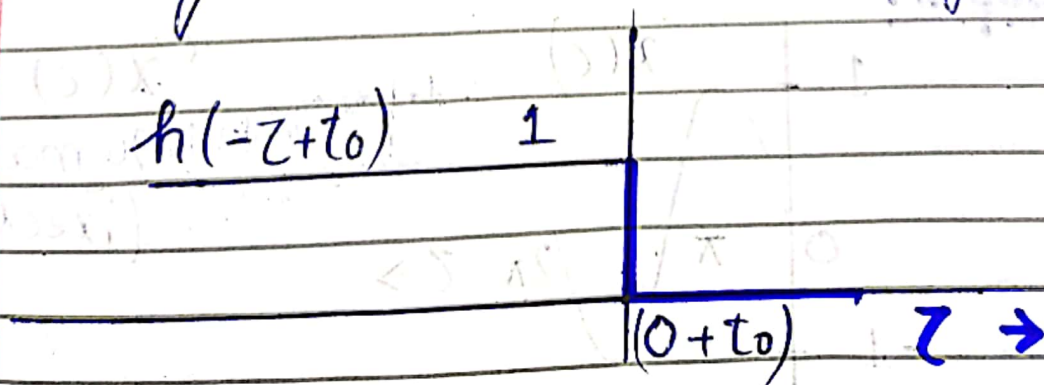
Step 2:-

"Rotate" $h(z)$ about vertical axis"



Step 3:-

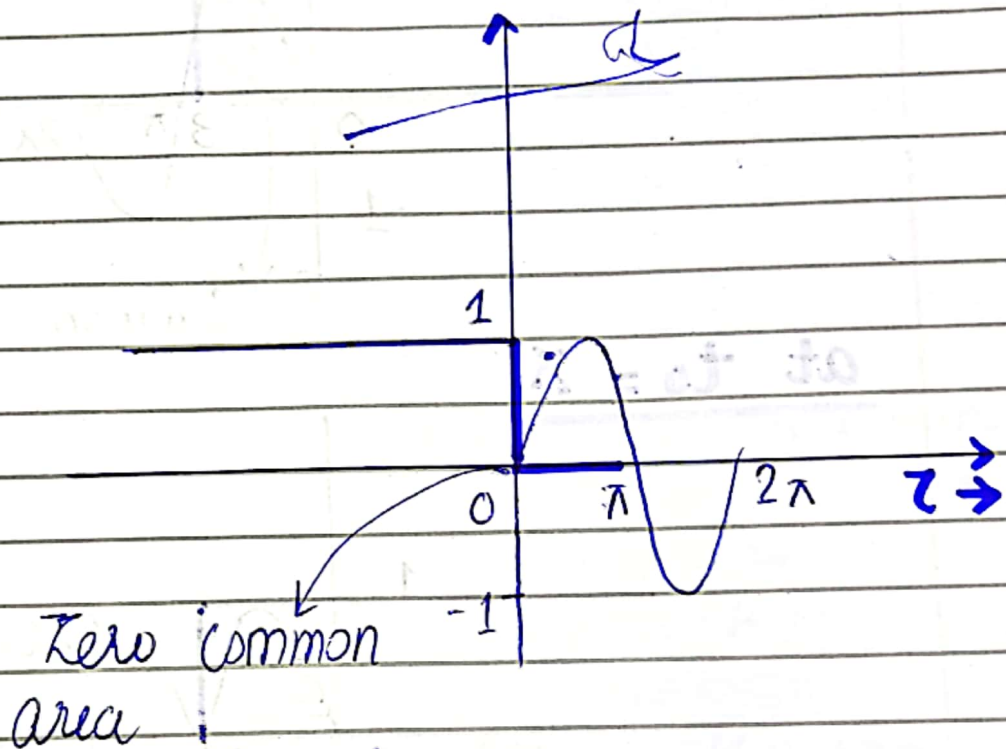
Shift the inverted frame $h(-z)$ along the z -axis by " t_0 ".



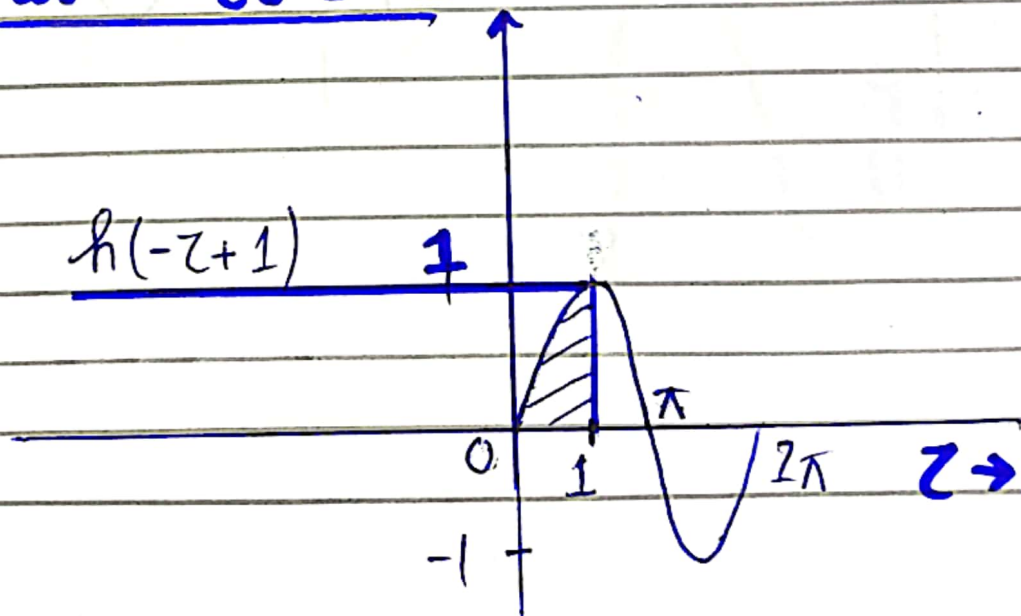
Step 4:-

Slide the $h(-\tau + t_0)$ near $x(\tau)$
at different " t_0 's value".

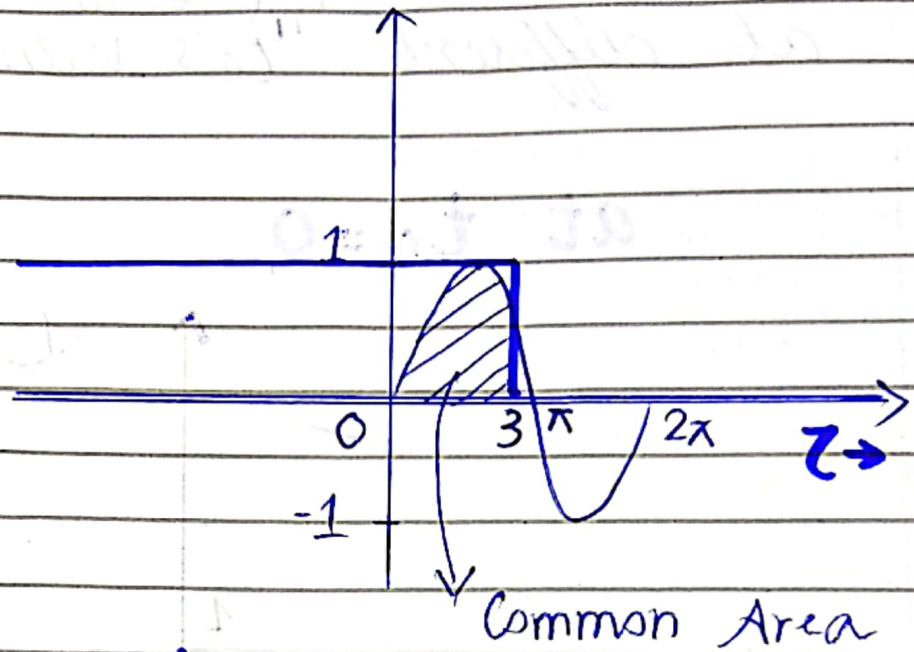
at $t_0 = 0$



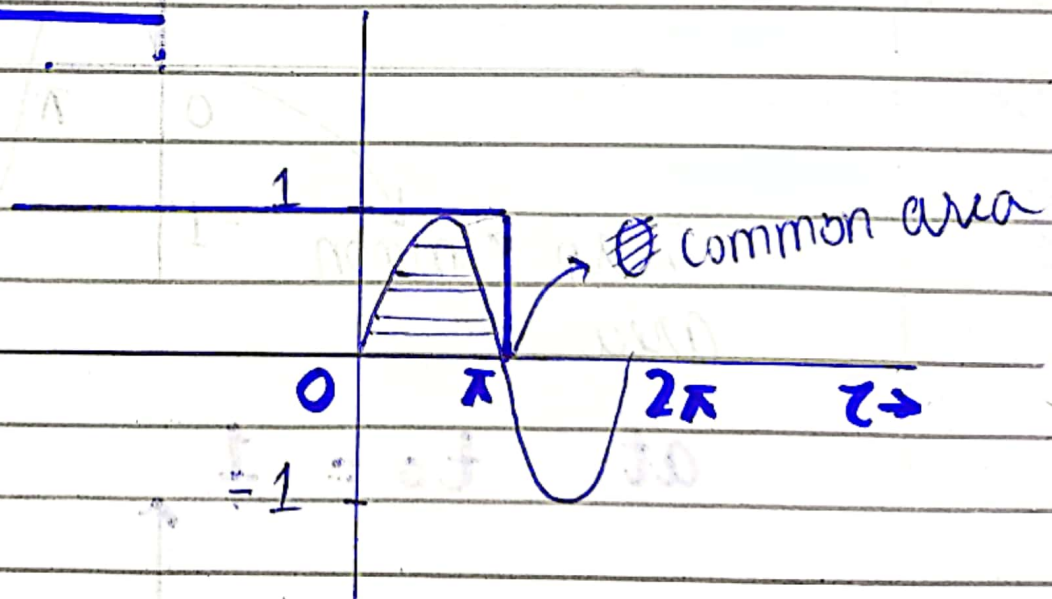
at $t_0 = 1$



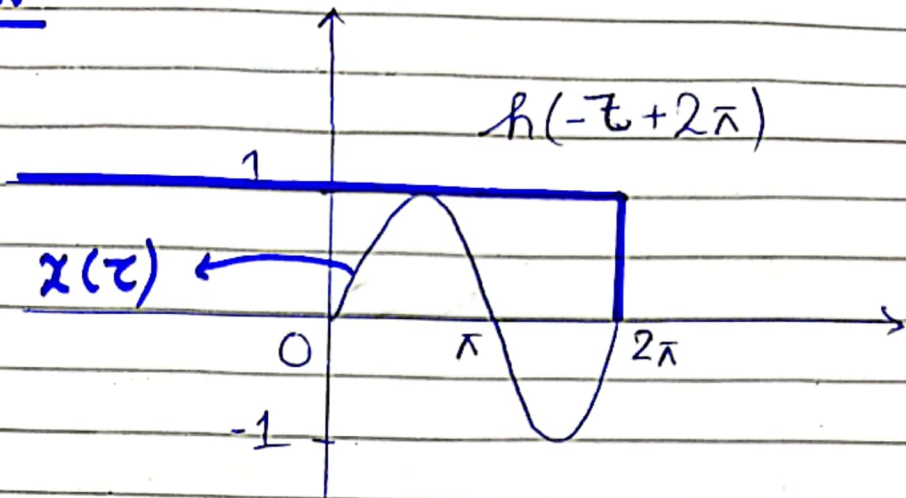
at $t_0 = 3$



at $t_0 = \pi$



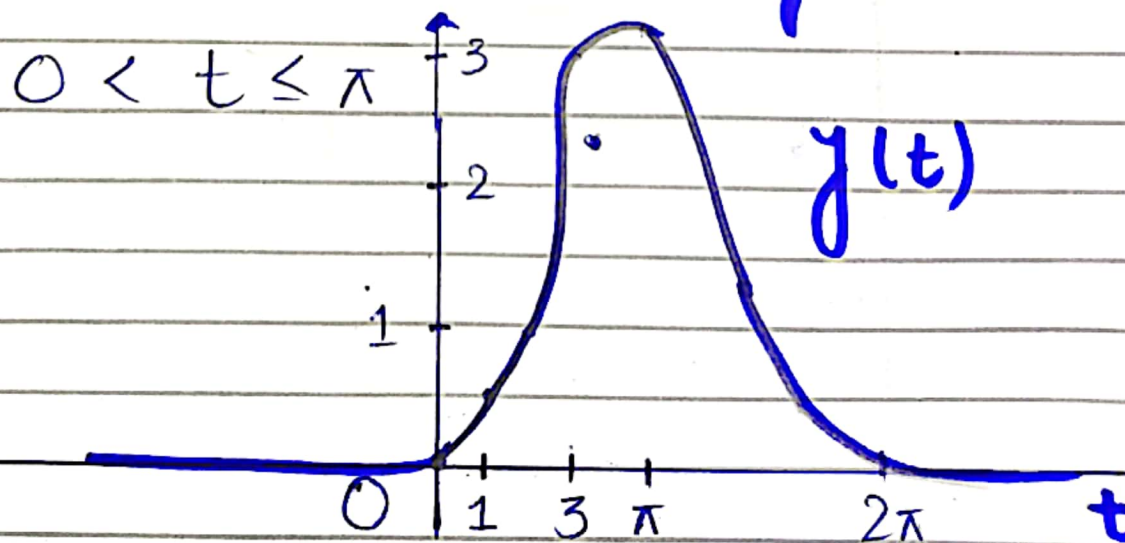
at $t = 2\pi$



Conclusion:-

- $t \leq 0$ there will be no common Area
- $t > 0$ Common Area but $t \leq \pi$

Common Area Range



Question #04

$$1) x(t) = e^{2(t-3)} u(t)$$

$$1) h(t) = e^{-t} u(t)$$

$$y(t) = x(t) \star h(t)$$

Solution :-

→ By Convolution integral :-

$$x_1(t) \star x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau$$

and

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \quad t \geq 0$$

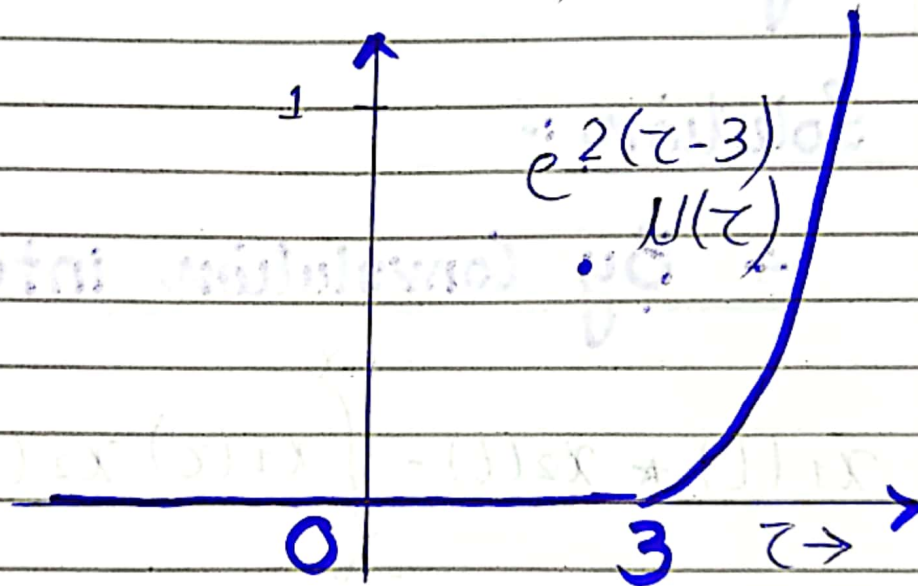
where;

$$x(t) = e^{2(t-3)} u(t)$$

"t" replaced by "τ"

⇒

$$x(\tau) = e^{2(\tau-3)} u(\tau)$$

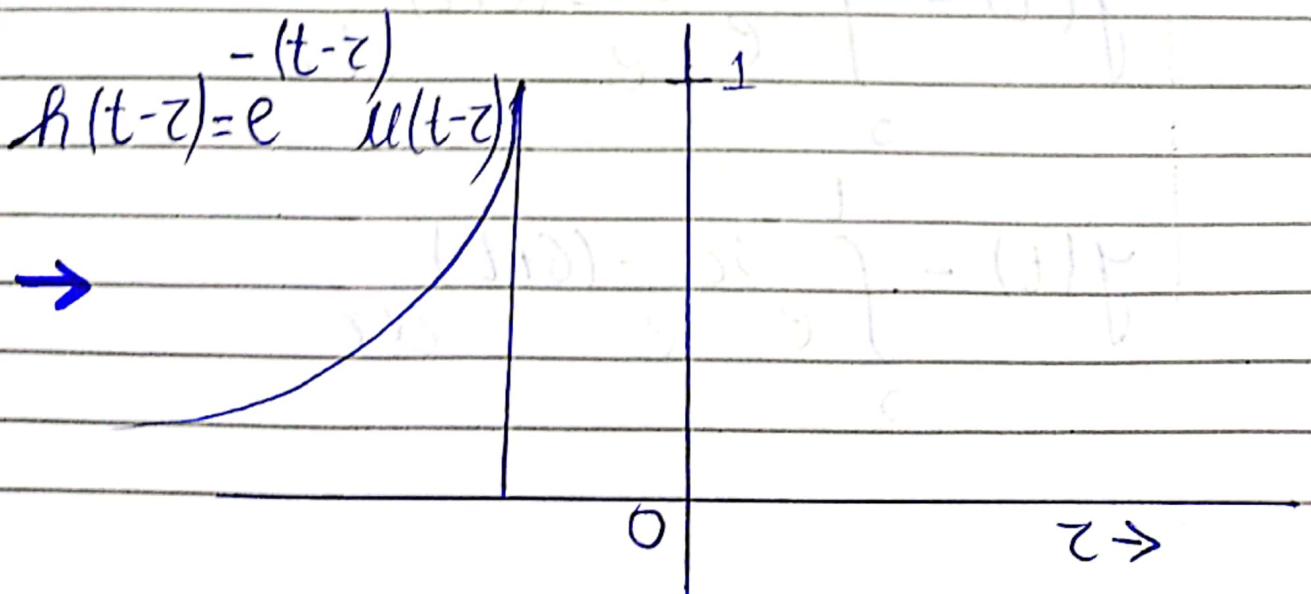
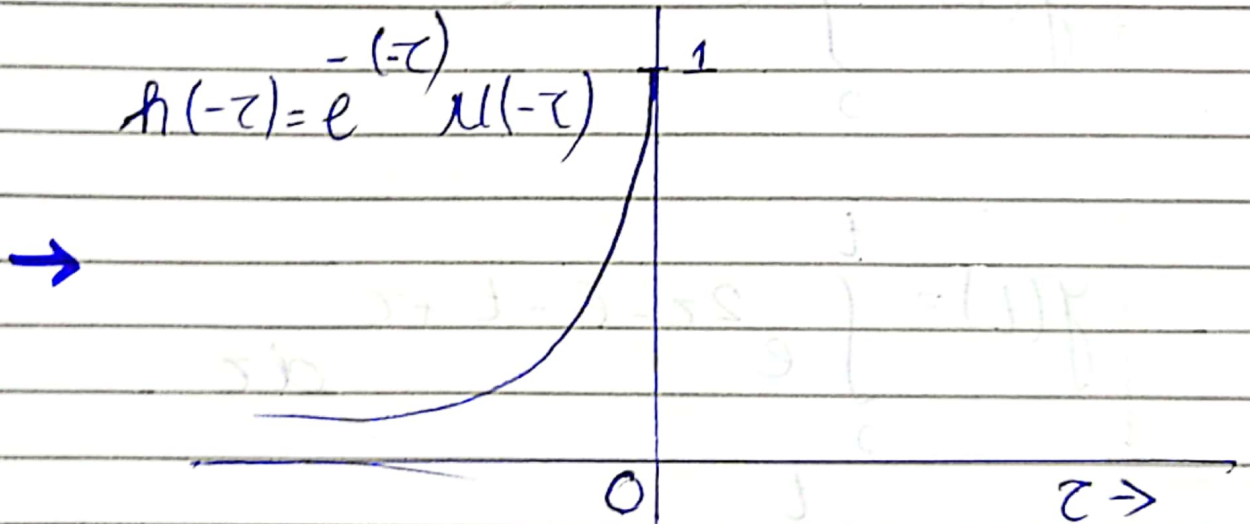
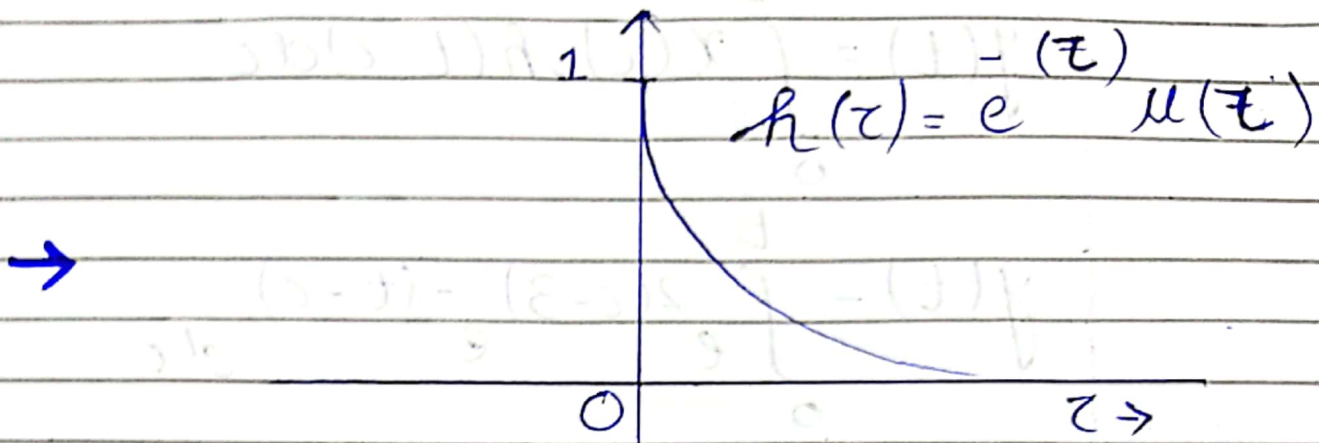


Similarly;

$$h(t) = e^{-t} u(t)$$

"t" replaced by (t-τ)

$$h(t-\tau) = e^{-(t-\tau)} u(t-\tau)$$



Now;

$$y(t) = \int_0^t x(\tau) h(t-\tau) d\tau$$

$$y(t) = \int_0^t e^{2(\tau-3)} e^{-(t-\tau)} d\tau$$

$$y(t) = \int_0^t e^{2\tau-6} e^{-t+\tau} d\tau$$

$$y(t) = \int_0^t e^{2\tau-6-t+\tau} d\tau$$

$$y(t) = \int_0^t e^{3\tau-(6+t)} d\tau$$

$$y(t) = \int_0^t e^{3\tau-(6+t)} d\tau$$

$$y(t) = e^{-(6+t)t} \int_0^t [e^{3\tau} d\tau]$$

$$y(t) = e^{-(6+t)t} \int_0^t [e^{3\tau} d\tau]$$

$$y(t) = e^{-(6+t)t} \left[\frac{e^{3\tau}}{3} \Big|_0^t \right]$$

$$y(t) = e^{-(6+t)t} \left[\frac{1}{3} (e^{3(t)} - e^{3(0)}) \right]$$

$$y(t) = e^{-(6+t)t} \left[\frac{1}{3} (e^{3t} - 1) \right]$$

$$y(t) = e^{-(6+t)t} \left[\frac{e^{3t}}{3} - \frac{1}{3} \right]$$

$$y(t) = \frac{e^{-(6+t)t}}{3} e^{3t} - \frac{e^{-(6+t)t}}{3}$$

$$y(t) = \frac{e^{-6+2t+3t} - e^{-(6+t)}}{3}$$

$$y(t) = \frac{e^{-6+2t} - e^{-(6+t)}}{3}$$

$$y(t) = \frac{1}{3} \left[e^{-6+2t} - e^{-(6+t)} \right]$$

Therefore;

$$y(t) = \frac{1}{3} \left[e^{-6+2t} - e^{-(6+t)} \right] u(t)$$

